

David Silveira

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Professional Summary

Dedicated and versatile Software Engineer with 5 years of experience in full-stack development, AI/ML, and DevOps. Proven expertise in building robust, scalable applications from the ground up, with a track record of driving projects from initial concept through to successful deployment and securing funding. Adept at leveraging modern technologies like React.js, Django, and Docker to create innovative solutions and streamline complex workflows.

Work Experience

Skaion Corporation | North Chelmsford

Software Engineer | February 2024 – January 2025

- Provided technical leadership and strategic direction for multi-million dollar government R&D project, collaborating with senior management to shape and refine core implementation plan
- Demonstrated technical agility by re-architecting core backend from Django framework to custom collection of Python scripts, successfully adapting solution to meet evolving project requirements
- Engineered full-stack solution, building dynamic React.js frontend and robust Python backend while leading DevOps process to ensure seamless packaging and deployment via Docker and Bash scripts
- Developed high-performance AI microservice with GPT4All, designing resource management system using Celery and Redis to optimize performance and enforce strict memory usage limits
- Cultivated strong relationships with 3 key DoD stakeholders communicating technical progress and proactively incorporating feedback.

Easy Covers Software | Wilmington

Founder/Lead Engineer | May 2023 – February 2024

- Drove end-to-end product lifecycle for advanced ATS platform, performing extensive user research with recruiters to design targeted features that achieved product-market fit
- Architected foundational infrastructure for the software, designing scalable microservices architecture and deploying secure reverse proxy to ensure high performance and reliability
- Established competitive advantage by engineering cutting-edge ASR pipeline and integrating generative AI to enable advanced speech recognition and speaker diarization
- Secured strategic partnerships and captivated investors by creating comprehensive business plan and showcasing software's value in high-stakes presentations
- Engineered entire full-stack platform from ground up, building robust Django backend and dynamic Next.js and React.js frontend to deliver fully functional product

Forum Systems | Needham

Software Engineer | February 2022 – May 2023

- Partnered with 2 executives to deploy machine learning-driven risk analysis feature, key innovation that secured follow-on funding from Blue Cross Blue Shield and enabled expansion of engineering team
- Engineered full-stack platform, developing robust Python/Django backend and dynamic React.js frontend to power suite of innovative features
- Utilized advanced machine learning models (including Transformer and GPT architectures) to perform sophisticated text analysis and enable precise risk score computation on PDF contracts
- Enhanced team productivity and streamlined onboarding process for 3 new developers by establishing 'dev-shortcuts' repository featuring 25+ productivity-enhancing shell script shortcuts
- Collaborated with small team of 4 to design and build innovative features, including NLP-based healthcare benefit encoding system and machine learning-powered product analysis engine

Kwak Laboratory | Amherst

Research Assistant | January 2020 – June 2021

- Engineered custom solution using R to automate transformation of unstructured text data into structured format, eliminating manual data handling
- Applied advanced data science techniques to political science study, uncovering key sentiment patterns from public reactions to election framing
- Synthesized complex text data into concise, interpretive reports, providing key insights that supported research findings of principal investigator
- Developed scalable data processing workflows that significantly reduced time required to analyze large datasets, accelerating research timelines

Skills

Web Development, Data Visualization, AI Integration, Shell Scripting, Text Analysis, Firewall Implementation, Data Structure Design, VPN Configuration, Cloud Service Management, Machine Learning, Security Scanning

Technologies

Programming Languages & Scripting: Python, JavaScript, Bash, R

AI, Machine Learning & NLP: GPT4All, NLP, ASR, Speaker Diarization, Word2Vec, Transformer Models, GPT Models, Kaldi, Hugging Face

Frameworks & Backend: Django, Flask, REST APIs, Celery, Nginx, Microservices

Frontend & UI/UX: React.js, Next.js, UX/UI Design, Figma, Material UI

DevOps & Infrastructure: DevOps, Docker, Containerization, Bash Scripts & Dockerfiles, Virtual Machines, Linux, Qubes OS, DHCP, DNS

Databases & Data: PostgreSQL, Redis, Microsoft Excel

Other Technologies & Tools: Open Source

Education

Bachelor of Science (BS) in Psychology, Neuroscience, and Computer Science | University of Massachusetts | 2021

Software Engineering Boot Camp | Columbia University | 2022